

Geometry of Recursive Knots:

Iterated Cables by Frame Offsets, Their Thickness and Ropelength

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Abstract

We study a concrete family of *recursive knots*: starting from a round circle K_0 , the curve K_d is the (p_d, q_d) torus pattern placed on the tube of radius r_d around K_{d-1} , using a rotation-minimising (Bishop) frame closed by a linear twist. The question of the research line is how much rope (length divided by thickness) such nested knots need. We prove three elementary facts for this family: K_d is an embedded smooth closed curve whenever $r_d < \text{thick}(K_{d-1})$ (Lemma 3.1); the closed Bishop frame has linking number $\text{round}(\text{Wr } K_{d-1})$ with its core, which identifies K_d as an explicit cable of K_{d-1} (Lemma 3.2); and $L(K_d) \leq p_d L(K_{d-1})(1 + r_d \kappa_{d-1}) + r_d(2\pi q_d + p_d |\alpha_d|)$ (Lemma 3.3). A lower bound on the thickness of K_d of the form $\min(\text{thick } K_{d-1} - r_d, c r_d \sin(\pi/p_d))$ is proved only for pairs of points in the same normal disc or over distant base points; the remaining local case is stated as a hypothesis (H_c) and checked numerically. Under (H_c) the ropelength bound satisfies a linear recursion whose solution grows geometrically, $\text{Rop}_d \leq 2\pi \Lambda^d$ with an explicit Λ . Numerically, explicit polygons with up to 8192 vertices give $L/\tau = 38.4, 155.6$ and 622.7 at depths 1, 2, 3 for the default pattern $(2, 3)$ and offset fraction $f = 1/2$, stable to 0.03% under doubling of the resolution. These are numerically verified upper bounds for explicit polygons, not certified smooth thicknesses; no stationarity, optimality or recursive flow law is claimed.

1 Introduction and status

The research line. Line RTK001 (“Geometry of recursive knots”) asks how nested knots—knots tied on the tube of other knots, recursively—can be represented as tubes without self-intersection, and how much length that costs. Its public record states one finding (a feasible upper bound at depth one) and an explicit scope restriction: neither stationarity nor optimality of any configuration is certified, and no general recursive flow law has been established. No manuscript of the line existed before this draft; the family below was fixed from the record and this document is a first written pass, with every number produced by the code in the reproducibility appendix.

What is done here. We (i) fix a definite family of recursive knots (Section 2); (ii) prove what is elementary about it—embeddedness, the framing it uses, and a length recursion—and isolate in one hypothesis what is not proved (Section 3); (iii) compute explicit polygonal configurations at depths 1 to 3, their polygonal thickness and ropelength, with a resolution check (Section 4); (iv) tabulate every claim with its status (Section 5). Lower bounds are only quoted from the literature: the crossing number of iterated cables is not known in general, so we derive no new lower bound.

2 Setting

2.1 Thickness and ropelength

Let $K \subset \mathbb{R}^3$ be a closed $C^{1,1}$ curve of length $L(K)$. Its *thickness* $\text{thick}(K)$ is the supremum of ρ such that the open tube of radius ρ about K is embedded (normal injectivity radius). Two equivalent descriptions are classical. Litherland, Simon, Durumeric and Rawdon [8] show

$$\text{thick}(K) = \min(\min\text{Rad}(K), \frac{1}{2} \text{dcsd}(K)), \quad (1)$$

where $\min\text{Rad}$ is the minimal radius of curvature and dcsd the *doubly critical self distance*, the minimal distance between pairs $x \neq y$ of K whose chord is perpendicular to K at both ends. Gonzalez and Maddocks [6] show $\text{thick}(K) = \inf_{x \neq y} \rho_{pt}(x, y)$, with $\rho_{pt}(x, y) = |x - y| / (2 \sin \angle(T_x, y - x))$ the radius of the circle through y tangent to K at x . The *ropelength* is $\text{Rop}(K) = L(K) / \text{thick}(K)$; it is scale invariant, $\text{Rop}(K) \geq 2\pi$ for every closed curve (total curvature $\geq 2\pi$ and $\kappa \leq 1/\text{thick}$), with equality only for the circle, and $\text{thick}(K) \leq \min\text{Rad}(K)$ always.

2.2 Frames, holonomy and the closing twist

Let K be a closed C^2 regular curve parametrised on $[0, 2\pi)$ (not necessarily by arclength), with unit tangent T and curvature κ . A *rotation-minimising* or Bishop frame [1] is an orthonormal frame (T, N_0, B_0) with $N'_0 = -\kappa_1 |K'| T$, $B'_0 = -\kappa_2 |K'| T$, $\kappa_1^2 + \kappa_2^2 = \kappa^2$; it is unique up to a constant rotation in the normal plane and is in general *not* periodic: after one turn the transported normal is rotated by a holonomy angle $\alpha \in (-\pi, \pi]$. We close it by the linear twist $\omega(t) = -\alpha t / 2\pi$, i.e. $N = \cos \omega N_0 + \sin \omega B_0$, $B = T \times N$. The closed frame satisfies

$$N' = -\kappa_1 |K'| T + \omega' B, \quad B' = -\kappa_2 |K'| T - \omega' N, \quad |\omega'| = \frac{|\alpha|}{2\pi} \leq \frac{1}{2}. \quad (2)$$

2.3 The family of recursive knots

Definition 2.1 (iterated cable by frame offsets). Let K_0 be the circle of radius $R_0 = 1$. For $d \geq 1$ choose coprime integers $p_d \geq 2$, $q_d \geq 1$, a radius $r_d > 0$ and a phase φ_d . Reparametrise K_{d-1} to period 2π by the linear change $t \mapsto t/p_{d-1}$ (so that K_{d-1} is traversed at parameter speed $|K'_{d-1}|$, not at unit speed), let (N_{d-1}, B_{d-1}) be its closed Bishop frame, and set for $t \in [0, 2\pi p_d)$

$$K_d(t) = K_{d-1}(t \bmod 2\pi) + r_d [\cos \theta_d(t) N_{d-1}(t) + \sin \theta_d(t) B_{d-1}(t)], \quad \theta_d(t) = \frac{q_d}{p_d} t + \varphi_d. \quad (3)$$

The default is $(p_d, q_d) = (2, 3)$, $\varphi_d = 0$ at every depth; r_d is a fraction f of the thickness of K_{d-1} (the guaranteed lower bound ρ_{d-1} in the theory, the measured polygonal value in the experiments).

K_1 is the (p_1, q_1) torus knot on the round torus of radii $(1, r_1)$; the knot type of K_d for $d \geq 2$ depends on the framing, which Lemma 3.2 makes explicit.

2.4 Polygonal measurements

Numerically a curve is a closed polygon $P = (x_0, \dots, x_{N-1})$. We use $L(P) = \sum |x_{i+1} - x_i|$; $\min\text{Rad}(P)$, the minimum circumradius of consecutive triples; $\text{dcsd}(P)$, the minimal distance $|x_i - x_j|$ over *doubly critical vertex pairs*, i.e. pairs where $|x_i - x_j|$ is a local minimum in i and in j (the diagonal $D_{ii} = 0$ is kept in the neighbourhood test, which excludes near-diagonal pairs); and the proxy

$$\tau(P) = \min(\min\text{Rad}(P), \frac{1}{2} \text{dcsd}(P)), \quad \text{Rop}(P) = L(P) / \tau(P). \quad (4)$$

As a cross-check we also compute $\tau_{pt}(P) = \min_{i \neq j} |x_j - x_i|^2 / (2|T_i \times (x_j - x_i)|)$ with central-difference tangents T_i . Neither τ nor τ_{pt} is a certified thickness of a smooth curve; they are the quantities of (1) evaluated on an explicit polygon, and their stability under doubling of N is the only evidence offered that they approximate the smooth values. Writhe is the Gauss double sum over pairs of edges (midpoint rule).

3 Results

Throughout, $K = K_{d-1}$ is a closed C^∞ regular curve of thickness $\tau = \text{thick}(K) > 0$ and maximal curvature $\kappa_{\max} \leq 1/\tau$, and K_d is built by (3) with $r = r_d$, $p = p_d$, $q = q_d$.

Lemma 3.1 (embedding). *[proved here] If $r \leq \tau/2$ then K_d is a regular C^∞ closed curve, injective on $[0, 2\pi p)$, contained in the torus $\partial\{x : \text{dist}(x, K) < r\}$.*

Proof. Smoothness: the Bishop frame of a C^∞ curve is C^∞ (it solves a linear ODE with smooth coefficients), and (3) is a smooth expression in it; periodicity of K_d in t with period $2\pi p$ holds because $\theta_d(t + 2\pi p) = \theta_d(t) + 2\pi q$ and the closed frame has period 2π . Regularity: differentiating (3) with (2),

$$K'_d = |K'| (1 - r(\kappa_1 \cos \theta + \kappa_2 \sin \theta)) T + r(\theta' + \omega') (-\sin \theta N + \cos \theta B), \quad (5)$$

and $|\kappa_1 \cos \theta + \kappa_2 \sin \theta| \leq \kappa \leq 1/\tau$, so the tangential coefficient is at least $|K'| (1 - r/\tau) \geq |K'|/2 > 0$. Injectivity: let $t_1 \neq t_2$ in $[0, 2\pi p)$. If $t_1 \not\equiv t_2 \pmod{2\pi}$, the two points lie in the normal discs of radius $r < \tau$ at distinct points of K ; by definition of thickness these discs are disjoint, so the points differ. If $t_1 \equiv t_2 \pmod{2\pi}$, then $t_1 - t_2 = 2\pi k$ with $0 < |k| < p$ and $\theta_d(t_1) - \theta_d(t_2) = 2\pi qk/p \notin 2\pi\mathbb{Z}$ because $\gcd(p, q) = 1$; the two points lie in the same normal disc at the same radius $r > 0$ but at different angles, hence differ. Containment in the torus is immediate from (3). $\square$

Lemma 3.2 (the framing used, and the knot type). *[proved here, from classical results] Let $\alpha \in (-\pi, \pi]$ be the holonomy of the Bishop frame of K and $n = \text{round}(\text{Wr } K)$ (the writhe is never a half-integer plus $1/2$ exactly when $\alpha \neq \pi$). Then the closed frame (N, B) is a framing of K with linking number $\text{Lk} = n$, $\alpha \equiv 2\pi \text{Wr } K \pmod{2\pi}$, and K_d is the $(p, q + pn)$ -cable of K (meridional winding measured against the Seifert framing). In particular K_1 is the torus knot $T(p_1, q_1)$.*

Proof. By the Călugăreanu–White–Fuller formula [4, 5, 10], $\text{Lk} = \text{Tw} + \text{Wr}$ for any framing. The Bishop frame has zero twist density; the closing rotation adds total twist $\text{Tw} = -\alpha/2\pi \in [-\frac{1}{2}, \frac{1}{2})$. Since $\text{Lk} \in \mathbb{Z}$, $\alpha \equiv 2\pi \text{Wr} \pmod{2\pi}$ and $\text{Lk} = \text{Wr} - \alpha/2\pi$ is the integer nearest to Wr . The pattern (3) winds q times around the meridian relative to the frame (N, B) while going p times along K ; relative to a framing of linking number 0 it winds $q + pn$ times, which is the definition of the $(p, q + pn)$ -cable [7, Ch. 1]. For K_0 a planar circle, $\text{Wr} = 0 = \alpha$ and K_1 is the standard torus knot. $\square$

Lemma 3.3 (length). *[proved here] Under the hypotheses of Lemma 3.1,*

$$L(K_d) \leq p L(K) (1 + r\kappa_{\max}) + r(2\pi q + p|\alpha|) \leq p L(K) (1 + r/\tau) + r(2\pi q + \pi p). \quad (6)$$

Proof. From (5), the two terms being orthogonal, $|K'_d| \leq |K'| (1 + r\kappa) + r(|\theta'| + |\omega'|) = |K'| (1 + r\kappa) + r(q/p + |\alpha|/2\pi)$. Integrate over $[0, 2\pi p)$: the first term gives $p \int_0^{2\pi} |K'| (1 + r\kappa) dt \leq p L(K) (1 + r\kappa_{\max})$, the second $r(2\pi q + p|\alpha|)$. Then use $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$. $\square$

Remark 3.4. The term $p|\alpha|r$ from the closing twist is absent from a naive count “ p copies plus q meridians”; it is at most πpr and is needed because the Bishop frame of a non-planar K does not close.

Thickness of K_d : what is proved and what is not. Write $s = 2r \sin(\pi/p)$ for the chord between adjacent strands in a normal disc of K (the p points $K_d(t + 2\pi k)$, $k = 0, \dots, p-1$, sit at the p equally spaced angles $\theta + 2\pi qk/p$, since $\gcd(p, q) = 1$).

Proposition 3.5 (two of the three cases). *[proved here] Under the hypotheses of Lemma 3.1: (a) two points of K_d in the same normal disc of K are at distance $\geq s$; (b) two points of K_d whose base points on K are at distance $\geq 2\tau$ are at distance $\geq 2(\tau - r)$.*

Proof. (a) They are at the same radius r and at angles differing by $2\pi qk/p$, $0 < k < p$, so their distance is $2r|\sin(\pi qk/p)| \geq 2r \sin(\pi/p) = s$ because $qk \not\equiv 0 \pmod{p}$. (b) Triangle inequality: $|K_d(t_1) - K_d(t_2)| \geq |K(t_1) - K(t_2)| - 2r$. $\square$

What Proposition 3.5 does not cover are pairs whose base points are close but distinct (the local geometry of p strands on a *bent* tube), and the curvature of K_d . For a straight tube the p strands are helices with pitch parameter $h = |K'|p/q$ and the antipodal pair ($p = 2$) is a local minimum of the strand distance exactly when $h \geq r$; the bending of K changes this by relative terms of order $r\kappa \leq 1/2$, which we have not controlled with explicit constants. We therefore state:

Hypothesis 3.6 (H_c). *[not proved; verified numerically for $c = 1/2$] There is a constant $c \in (0, 1]$, depending only on (p, q) and on $f = r/\tau \leq 1/2$, such that*

$$\text{thick}(K_d) \geq \rho_d := \min(\tau - r, cr \sin(\pi/p)). \quad (7)$$

The form of (7) is what Proposition 3.5 suggests ($cr \sin(\pi/p) = \frac{c}{2}s$ is half the strand chord, times a safety factor), and $c = 1$ would be sharp for $p = 2$ on a straight tube. In the experiments of Section 4, $\tau(K_d)/\rho_d \geq 0.83$ with $c = 1/2$ in all 9 default levels with $f \leq 1/2$, and $\tau(K_d) = r_d$ to four digits in 8 of them.

Corollary 3.7 (recursion for the ropelength bound). *[proved here, conditional on H_c] Fix (p, q) , $f \leq 1/2$ and c as in (H_c) , and build the family with $\rho_0 = 1$, $r_d = f\rho_{d-1}$. Then $\rho_d = \gamma^d$ and $L(K_d) \leq 2\pi A^d$ with*

$$\gamma = \min(1 - f, cf \sin(\pi/p)), \quad A = p(1 + f) + f(q + \frac{p}{2}), \quad (8)$$

hence $\text{Rop}(K_d) \leq L(K_d)/\rho_d \leq 2\pi\Lambda^d$ with $\Lambda = A/\gamma$. For the default $(2, 3)$, $f = 1/2$, $c = 1/2$: $\gamma = 0.250$, $A = 5.00$, $\Lambda = 20.0$.

Proof. By (H_c) and $r_d = f\rho_{d-1}$, $\rho_d = \rho_{d-1} \min(1 - f, cf \sin(\pi/p)) = \gamma\rho_{d-1}$. By Lemma 3.3 with $\tau \geq \rho_{d-1}$, $L(K_d) \leq pL(K_{d-1})(1 + f) + f\rho_{d-1}(2\pi q + \pi p)$; since $\rho_{d-1} \leq \text{thick}(K_{d-1}) \leq L(K_{d-1})/2\pi$, this is $\leq L(K_{d-1})[p(1 + f) + f(q + p/2)] = AL(K_{d-1})$, and $L(K_0) = 2\pi$. $\square$

Remark 3.8 (on the constants). Λ is generous: Lemma 3.3 bounds $|K'_d|$ by the sum of the norms of two orthogonal vectors, uses $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$, and (H_c) with $c = 1/2$ is twice weaker than the observed $\tau_d \approx r_d$. Numerically the growth factor per depth is about 4.06–4.00 (Table 1) against $\Lambda = 20.0$. We make no claim that any configuration in the family is stationary or optimal for ropelength, and none that the recursion is a “flow law” for tight recursive knots: it bounds one explicit construction.

Remark 3.9 (lower bounds, quoted). [literature] Buck and Simon [2] prove $\text{Rop}(K) \geq c_0 \text{Cr}(K)^{3/4}$ for a universal $c_0 > 0$, and Cantarella, Kusner and Sullivan [3] prove that ropelength minimisers exist in every knot and link type, are $C^{1,1}$, and satisfy further lower bounds obtained from cone surfaces. To turn these into a lower bound for K_d one needs $\text{Cr}(K_d)$; for the iterated cables of Lemma 3.2 the crossing number is not known in general (crossing numbers of satellites are an open problem), so we derive no lower bound here. For $K_1 = T(2, 3)$ the best known numerical ropelength is about 32.7 [3, 9]; our depth-one polygon at $f = 1/2$ gives 38.4, i.e. the construction is not close to tight even at depth one.

Conjecture 3.10 (what the numbers suggest). [conjectural] For the default family with $f \leq 1/2$, $\text{thick}(K_d) = r_d$ exactly whenever the antipodal pair in a normal disc is the doubly critical pair (observed for $f \leq 0.35$ at all depths and for $f = 1/2$ at $d \geq 2$), and $\text{Rop}(K_d)$ grows like $\Theta(\Lambda_*^d)$ with $\Lambda_* = \lim L(K_d)/(fL(K_{d-1})) \approx 4.00$. Neither statement is proved.

4 Experiments

Protocol. `experiments/recursive_knots.py` builds the polygon K_0 with N_0 vertices, computes the closed Bishop frame by discrete parallel transport (rotation taking T_i to T_{i+1}) with the holonomy removed by the linear twist, and produces K_d with $p_d N_{d-1}$ vertices by (3) evaluated at the vertices (so no interpolation is used; the parameter t is the vertex index). At each depth it measures L , minRad , dcsd , τ , τ_{pt} , Rop , the writhe and the holonomy, and sets $r_{d+1} = f \tau(K_d)$. Grid: $(p, q) = (2, 3)$ with $f \in \{0.25, 0.35, 0.5, 0.6\}$, and $(3, 2)$, $(2, 5)$ with $f = 0.5$; depths 1–3; two resolutions, $N_0 \in \{512, 1024\}$ ($\{256, 512\}$ for $(3, 2)$, whose depth-3 polygon has $27N_0$ vertices). The value $f = 0.6$ lies outside the hypothesis $r \leq \tau/2$ and is included only as a numerical probe. The computation is deterministic; total time 4.7 CPU-minutes.

Predefined criteria. (C1) resolution: Rop and τ change by less than 1% between N_0 and $2N_0$; (C2) $L(K_d) \leq$ the right-hand side of (6) evaluated with the measured $L(K_{d-1})$, $\tau(K_{d-1})$ and α_d ; (C3) $\tau(K_d) \geq \rho_d$ of (7) with $c = 1/2$ and the measured $\tau(K_{d-1})$, for $f \leq 1/2$; (C4) τ_{pt} agrees with τ to within 1%.

Results. Table 1 gives the default family, Table 2 the two other patterns, Table 3 the resolution check and Table 4 the comparison with Corollary 3.7; Figure 1 shows the curves and Figure 2 the growth. All four criteria hold: (C1) the maximal relative change is 0.03% for Rop and 0.03% for τ over all 12 chains; (C2) holds in 36 of 36 levels; (C3) holds in all 9 default levels with $f \leq 1/2$, with margin $\tau_d/\rho_d \geq 0.83$ (and ≥ 0.56 including the other patterns); (C4) the maximal discrepancy is 0.01%. In every level τ is determined by $\text{dcsd}/2$, never by minRad (36 of 36).

Three further observations. First, for $f \leq 0.35$ the measured $\tau(K_d)$ equals r_d to four digits at every depth (the antipodal strands in a normal disc are the doubly critical pair), so $\text{Rop}_d = L_d/r_d$ and the recursion $\tau_d = f\tau_{d-1}$ is exact; for $f = 1/2$ at depth one $\tau_1 = 0.4158 < r_1 = 0.5$, the bending of the base torus bringing inner strands closer, while at depths 2, 3 again $\tau_d = r_d$. Second, the writhe of $K_1 = T(2, 3)$ crosses the half-integer 3.5 inside the grid ($\text{Wr} = 3.127, 3.257, 3.518, 3.710$ for $f = 0.25, 0.35, 0.5, 0.6$), so by Lemma 3.2 the frame linking number is 3 for $f \leq 0.35$ and 4 for $f \geq 0.5$ and the depth-two knot type changes across the grid: K_2 is the $(2, 9)$ -cable of the trefoil for $f = 0.25$ but the $(2, 11)$ -cable for $f = 0.5$. The measured α_d agrees with $2\pi \text{frac}(\text{Wr } K_{d-1})$ as Lemma 3.2 requires (e.g. $\alpha_2 = -3.027$ for $f = 0.5$). Third, the best depth-one value on the grid is $f = 0.5$ ($\text{Rop}_1 = 38.4$), the best depth-three value also ($\text{Rop}_3 = 622.7$); $f = 0.6$ is worse at every depth, consistent with the crossover $1 - f \approx cf$ in (8), but we did not optimise f and make no optimality claim.

Table 1: Default family $(p, q) = (2, 3)$, finest resolution ($N_0 = 1024$). $r_d = f\tau_{d-1}$; ρ_d from (7) with $c = 1/2$; slope = $q + pn$ of Lemma 3.2. All lengths in units of R_0 .

f	d	N	r_d	L	minRad	dcsd / 2	τ	L/τ	ρ_d	Wr	slope
0.25	1	2048	0.2500	13.444	0.860	0.2500	0.2500	53.8	0.2500	3.13	3
0.25	2	4096	0.0625	26.911	0.818	0.0625	0.0625	430.6	0.0625	15.25	9
0.25	3	8192	0.0156	53.822	0.808	0.0156	0.0156	3444.6	0.0156	63.50	33
0.35	1	2048	0.3500	14.271	0.797	0.3500	0.3500	40.8	0.3500	3.26	3
0.35	2	4096	0.1225	28.608	0.705	0.1225	0.1225	233.5	0.1225	15.49	9
0.35	3	8192	0.0429	57.219	0.675	0.0429	0.0429	1334.6	0.0429	63.98	33
0.5	1	2048	0.5000	15.949	0.715	0.4158	0.4158	38.4	0.5000	3.52	3
0.5	2	4096	0.2079	32.349	0.639	0.2079	0.2079	155.6	0.2079	17.99	11
0.5	3	8192	0.1040	64.730	0.544	0.1040	0.1040	622.7	0.1040	74.96	39
0.6	1	2048	0.6000	17.307	0.688	0.3401	0.3401	50.9	0.4000	3.71	3
0.6	2	4096	0.2041	34.941	0.571	0.1367	0.1367	255.6	0.1360	18.36	11
0.6	3	8192	0.0820	69.893	0.492	0.0820	0.0820	852.1	0.0547	75.70	39

Table 2: Other patterns, $f = 0.5$, finest resolution ($N_0 = 512$ for $(3, 2)$, 1024 for $(2, 5)$).

(p, q)	d	N	r_d	L	τ	L/τ	ρ_d	Wr	slope
(3, 2)	1	1536	0.5000	20.000	0.4191	47.7	0.4330	3.90	2
(3, 2)	2	4608	0.2096	60.087	0.1813	331.4	0.1815	39.66	14
(3, 2)	3	13824	0.0906	180.270	0.0785	2296.5	0.0785	362.97	122
(2, 5)	1	2048	0.5000	20.420	0.2822	72.4	0.5000	6.87	5
(2, 5)	2	4096	0.1411	41.116	0.1411	291.4	0.1411	32.66	19
(2, 5)	3	8192	0.0705	82.273	0.0705	1166.2	0.0705	136.30	71

5 Claims and their status

Claim	Status
K_d is embedded, smooth and regular when $r_d \leq \text{thick}(K_{d-1})/2$ (Lemma 3.1).	proved here
The closed Bishop frame has $\text{Lk} = \text{round}(\text{Wr})$; K_d is the $(p, q + pn)$ -cable of K_{d-1} (Lemma 3.2).	proved here from classical results
$L(K_d) \leq pL(K_{d-1})(1 + r\kappa_{\max}) + r(2\pi q + p \alpha)$ (Lemma 3.3).	proved here
Same-disc pairs at distance $\geq 2r \sin(\pi/p)$; far pairs at distance $\geq 2(\tau - r)$ (Prop. 3.5).	proved here
$\text{thick}(K_d) \geq \min(\tau - r, cr \sin(\pi/p))$, $c = 1/2$ (Hypothesis H_c).	not proved; verified numerically in 9 levels, margin ≥ 0.83
$\text{Rop}(K_d) \leq 2\pi\Lambda^d$, $\Lambda = A/\gamma$ (Cor. 3.7).	proved conditional on H_c
Explicit polygons with $L/\tau = 38.4, 155.6, 622.7$ at $d = 1, 2, 3$ ($f = 0.5$), stable to 0.03%.	computationally verified (polygonal proxy, not certified smooth thickness)
Knot type of K_2 changes between $f = 0.35$ and $f = 0.5$ (writhe crosses 3.5).	computationally verified (Wr to 10^{-3}), consequence of Lemma 3.2
$\text{Rop} \geq c_0 \text{Cr}^{3/4}$; existence and regularity of tight knots.	literature [2, 3]
$\text{thick}(K_d) = r_d$ exactly for small f ; growth rate $\Lambda_* \approx 4.00$.	conjectural
Stationarity or optimality of any K_d ; a recursive flow law for tight recursive knots.	not claimed

Table 3: Resolution check (criterion C1): L/τ at the coarse and fine resolutions and relative change.

(p, q)	f	d	N coarse	L/τ	N fine	L/τ	change (%)
(2, 3)	0.25	1	1024	53.78	2048	53.78	0.001
(2, 3)	0.25	2	2048	430.57	4096	430.57	0.001
(2, 3)	0.25	3	4096	3444.61	8192	3444.63	0.001
(2, 3)	0.35	1	1024	40.77	2048	40.77	0.001
(2, 3)	0.35	2	2048	233.54	4096	233.54	0.001
(2, 3)	0.35	3	4096	1334.55	8192	1334.56	0.001
(2, 3)	0.5	1	1024	38.36	2048	38.36	0.001
(2, 3)	0.5	2	2048	155.59	4096	155.59	0.001
(2, 3)	0.5	3	4096	622.67	8192	622.67	0.001
(2, 3)	0.6	1	1024	50.88	2048	50.89	0.003
(2, 3)	0.6	2	2048	255.52	4096	255.60	0.029
(2, 3)	0.6	3	4096	851.87	8192	852.13	0.029
(3, 2)	0.5	1	768	47.71	1536	47.72	0.004
(3, 2)	0.5	2	2304	331.42	4608	331.45	0.007
(3, 2)	0.5	3	6912	2296.29	13824	2296.46	0.007
(2, 5)	0.5	1	1024	72.36	2048	72.36	0.003
(2, 5)	0.5	2	2048	291.40	4096	291.41	0.003
(2, 5)	0.5	3	4096	1166.18	8192	1166.21	0.003

Table 4: Measured L/τ (numerical chain, $r_d = f\tau_{d-1}$) against the conditional bound $2\pi\Lambda^d$ of Corollary 3.7 (theoretical chain, $r_d = f\rho_{d-1}$, $c = 1/2$). The two chains differ in r_d ; the comparison is indicative only.

f	γ	A	Λ	$d = 1$	$d = 2$	$d = 3$
0.25	0.125	3.50	28.0	53.8 / 176	430.6 / 4926	3444.6 / 137928
0.35	0.175	4.10	23.4	40.8 / 147	233.5 / 3449	1334.6 / 80801
0.5	0.250	5.00	20.0	38.4 / 126	155.6 / 2513	622.7 / 50265

6 Limitations

(1) The thickness lower bound is proved only in two of three cases; the local case on a bent tube and the curvature of K_d are covered by (H_c) , which is checked numerically for one value of c on a small grid. (2) The polygonal τ of (4) uses vertex pairs, not segment pairs, and a local-minimum test; agreement with τ_{pt} to 0.01% and stability under doubling of N are evidence, not certification. Certified bounds would require interval arithmetic on a $C^{1,1}$ interpolant. (3) r_d is chosen as a fixed fraction of the measured thickness; nothing is optimised, and the family (pattern angle linear in the parameter t , not in arclength; phase $\varphi_d = 0$) is one of many. (4) The comparison of Table 4 mixes two chains with different r_d . (5) Writhe and holonomy are midpoint-rule sums; their agreement to 10^{-3} under doubling of N is the only accuracy check, and the frame linking number for $f = 0.5$ ($Wr = 3.518$) is decided by a margin of 0.02. (6) No lower bound specific to K_d is derived, because $Cr(K_d)$ is not known.

7 Next steps

(1) Prove (H_c) : control the strand distance on a bent tube and the curvature of K_d with explicit constants, using (5) and its derivative; the antipodal-pair analysis for $p = 2$ ($h \geq r$) is the natural start. (2) Replace the polygonal proxy by a certified computation (interval arithmetic on minRad and dcsd of a piecewise-circular interpolant). (3) A constrained length-shrinking step at fixed $\tau \geq 1$ for $d = 1, 2$, reported as uncertified numerical improvement, to see how far the family is from tight configurations. (4) Optimise f_d per depth and the pattern angle in arclength rather than in t . (5) Compute the crossing numbers of the specific cables identified in

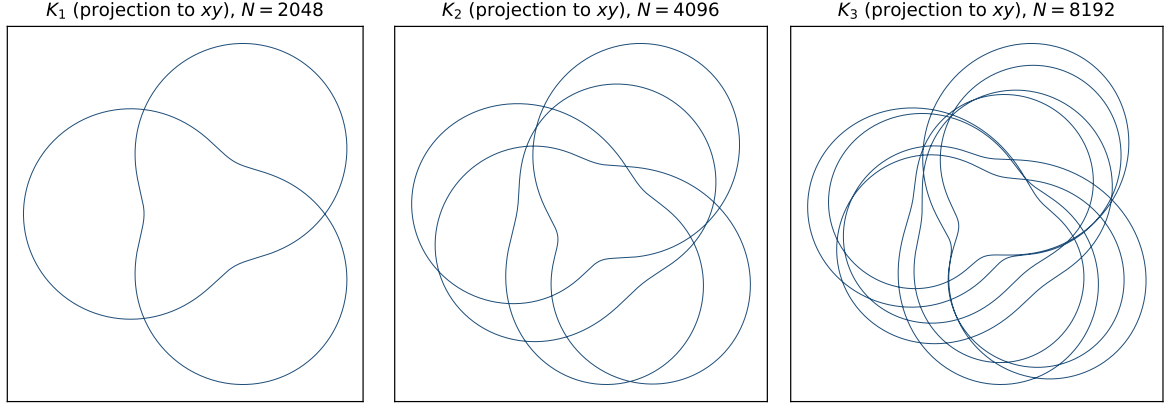


Figure 1: Projections of K_1, K_2, K_3 for $(2, 3)$, $f = 0.5$.

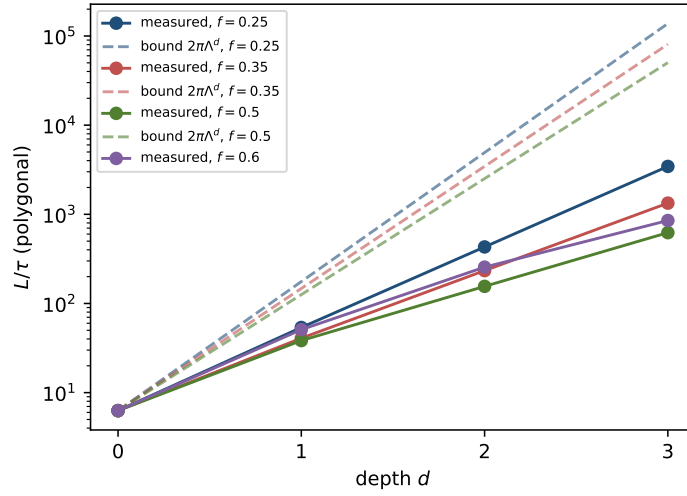


Figure 2: Polygonal L/τ against depth for the default family (solid) and the conditional bound $2\pi\Lambda^d$ (dashed, $f \leq 1/2$).

Lemma 3.2 where known, to attach a Buck–Simon lower bound to each explicit configuration.

A Reproducibility

`python3 experiments/recursive_knots.py` writes `results/results.json`, `results/tables.md` and the figures (`-fast` halves both resolutions); `python3 experiments/make_numbers.py` writes `manuscript/numbers.tex` and the table bodies, and `manuscript/build.sh` runs both and `latexmk`. Every number in this text is a macro from `numbers.tex`. Reference run: seed 20260930 (bookkeeping only; the computation is deterministic), Python 3.11.15, NumPy 2.4.6, 281 s of wall time on shared cores. Parameters: $R_0 = 1$, $N_0 \in \{512, 1024\}$, depth 3, $f \in \{0.25, 0.35, 0.5, 0.6\}$, patterns $(2, 3)$, $(3, 2)$, $(2, 5)$, $\varphi_d = 0$. Discrete Bishop frame: parallel transport by the rotation taking T_i to T_{i+1} (Rodrigues formula), initial normal orthogonal to T_0 , holonomy measured as the signed angle between the transported and the initial normal, removed by the twist $-\alpha i/N$.

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